

Global Wealth, Education and Population: A Data Visualisation Analysis

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1 PART 1. ANALYTICS

I agree to abide by the expectations as to my conduct, as described in the academic honesty and integrity statement.

1.1 Research Questions

In the current global geopolitical scenario, exploring the relationship between national wealth and education is relevant and timely. While comparing this relation among different economies of the world, population plays a crucial role, because a small wealthy country achieving 90% secondary school enrolment is quite different and incomparable to large countries achieving the same if population size is not considered. Hence, a population dataset was included to strengthen the analysis and findings.

Q1: Analyse the relation between wealth and education. Are there detectable trends?

Q2: Does government investment in education drive future wealth growth, or does wealth drive educational investment?

This question asks about the direction of the cause-effect relationship between global wealth and education, whether spending on education results in wealth growth or wealthy countries spend on education because they can afford it. The analysis of this question helps understand what affects investment decisions in global economies. If spending drives wealth, it justifies education as a development investment, whereas if wealth drives spending, it suggests education improvement is a byproduct of economic growth.

Q3: Are low-income countries catching up with wealthier nations in terms of education over time, and is this consistent across all parts of the world?

This question answers whether the gap between rich and poor countries is closing or widening over time. By analysing across the world economies, we can determine if the change is consistent across all regions or is it specific. Q1 shows the global picture at a given moment but does not reveal whether the gap is improving over time, whereas Q3 directly addresses this across different income groups and regions.

1.2 Data and Datasets

1.2.1 Data for Q1

Primary dataset: IWI Wealth Index (Global Data Lab, 133 countries, 1992-2023) [2]. This analysis uses the national-level IWI score because it measures actual household asset ownership (electricity, clean water, appliances etc.), rather than just national income. It reflects the living standard of ordinary people, which is more directly comparable across countries than national wealth.

Secondary dataset: World Education Data (World Bank via Kaggle, 256 countries, 1999-2023) [6]. This analysis uses secondary school enrolment percentage as primary education metric because literacy rate has 68% missing data (not enough reliable information) and primary enrolment is almost similar for most countries. Secondary enrolment has 88.6% data after merging datasets and has the strongest correlation with wealth ($r=0.876$).

Additional dataset: World Bank Population Data (World Bank, 266 countries, 1960-2024) [5]. Total population is used as bubble

size in Q1 visualisation, and also income group classification from this dataset's metadata.

Limitations: The merged dataset covers 126 countries only, which is less than half of 256 in the education dataset because the IWI only covers 133 countries. This means very small nations are underrepresented.

1.2.2 Data for Q2

This analysis uses `gov_exp_gdp` (government education expenditure as % of GDP) from the education dataset, and `iwi_score` from the IWI dataset, because government expenditure as % of GDP is a standard measure of commitment to education, comparable across countries regardless of economic size. It answers the question whether spending in a year predicts IWI growth for the next 5 years.

Limitations: `gov_exp_gdp` is 25.5% missing in merged dataset, which reduces the sample. Also, sources of education funding can include private spending, NGO investment, etc. other than government spending, so it can only establish predictive relationship rather than establishing the cause.

1.2.3 Data for Q3

This analysis uses `secondary_enrol` over time, grouped by income classification from the World Bank population metadata, and by continent from the IWI dataset. Tracking secondary enrolment over the years within each income group directly shows whether the gap is closing.

Limitations: Income group classification changes over time but it is used throughout for simpler and uniform analysis which might not represent that countries upgrade their income classification over the years.

1.3 Dataset Relationships

The IWI and education datasets are joined on ISO 3-letter country code and year, producing a merged dataset of 1504 rows covering 126 countries between 2000 and 2021. The population dataset is joined using the same ISO country code and year (all 3 datasets use the World Bank ISO standard, so the join is clean).

The IWI measures wealth from the bottom, i.e. household assets while the education dataset measures human capital investment at the national level, and the population dataset provides demographic scale and income classification. Together, these datasets determine whether wealth and education correlate, how many people are affected, what governments are doing about it, and whether the relationship is improving over time. Using a single dataset does not answer these questions alone.

The IWI has data from 1992-2023 but has sparse values before 1999. Education dataset is from 1999-2023, and population dataset covers 1960-2024. The common duration with strong data values is 2000-2021 (22 years), which is the time period this analysis uses to identify meaningful trends and relationships between global wealth, education and population.

Limitations: The IWI is the smallest dataset with 133 countries and determines how many countries appear in the final merged datasets. Many high income countries like USA, UK, Germany are missing from IWI because it was designed primarily for measuring

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development in lower and middle income countries. This means that this analysis focuses on developing and emerging countries.

2 PART 2. DESIGN AND DISCUSSION

Three visualisations are designed to answer Q1, Q2 and Q3 respectively, each exploring the wealth-education-population relationship from different analytical angles.

2.1 Visualisation 1: Wealth vs. Education Over Time (Q1)

IWI score and enrolment are quantitative variables. To represent quantitative variables, channels like Position, Length, Area or Colour are used. To depict the correlation between quantities like IWI score and enrolment, Position channel works best for x-y data points. In this case, line or bar charts cannot be used as they only represent one variable using length, not position, whereas a bubble-scatter plot works best here.

In the graph, population is represented as bubble size, because the wealth of a country can be affected by population size. More people can mean less wealth per person, and vice versa. Hence population becomes a third analytical dimension in the wealth-education comparison.

There are 126 countries in the final merged dataset, but we cannot assign colours to individual countries as it may create visual confusion, hence countries are grouped by continent and assigned a colour, giving 4 distinct colours. Meanwhile, income group cannot be used for colour as Q1 is about geographic pattern rather than economic classification.

A static chart represents 22-year period in one flat view and does not show how the relationship evolved over time. A year slider is therefore added so viewer can observe trajectories of countries year-by-year, like which countries improved, which stagnated and which declined, directly supporting Q1's question of whether detectable trends exist.

The strength of this design is that it shows 3 variables plus time simultaneously in a single view. The weakness is that bubble area is a less accurate channel than position, so size comparisons between countries becomes difficult. This design is inspired by Gapminder [1], focused on the wealth-education-population relationship rather than GDP and life expectancy (see Fig. 1).

2.2 Visualisation 2: Education Investment vs. Wealth Growth (Q2)

A scatter plot is used for this visualisation because Q2 analyses the cause-effect relationship between government education spending and national wealth. Government education expenditure (% of GDP) is plotted on the x-axis and IWI wealth growth on the y-axis, as position channel works best for showing correlation between two quantitative variables. Line or bar charts cannot be used as they do not show the relationship between two quantitative variables simultaneously.

Income group is used for colour because Q2 specifically asks whether the spending-wealth relationship differs across economic tiers. Through the data, we can see that in low income countries, higher spending shows a weak positive lagged effect on wealth growth, whereas in high income countries the effect is negative because they already have high education levels. Colouring by continent would not show this difference because a continent contains both low and lower-middle income countries mixed together, which will hide the economic disparity rather than depict them.

A scatter plot can show the spending-wealth relationship at a single point in time. But to answer whether spending in one year predicts wealth growth over the next 5 years, the x-axis shows government spending in year Y and the y-axis shows IWI growth from year Y to Y+5. This 5-year lag is precalculated during data preprocessing and explained in the chart subtitle so that it is clear what the axes represent.

This visualisation adds a new analytical dimension that Visualisation 1 could not show. It showed the correlation between wealth and education with respect to population across countries but did not determine which is the cause and which is the effect. Visualisation 2 addresses this by showing whether higher spending leads to higher wealth growth, or wealthier countries spend more just because they can afford it.

The strength of this design is that it directly addresses the cause-effect direction of the relationship. The weakness is that correlation does not prove causation, plus some missing government spending data has reduced the sample size (see Fig. 2).

2.3 Visualisation 3: Educational Convergence Across Income Groups (Q3)

A line chart is used for this visualisation because Q3 analyses the trend of secondary enrolment across different income groups over time. A line chart works best here as it shows one quantitative variable against time, and multiple lines can be compared simultaneously without visual crowding. A stacked bar chart could show multiple groups but would become crowded and harder to read, whereas individual lines clearly show which income groups improved, which stagnated and which declined over the years.

Year is plotted on the x-axis and secondary school enrolment percentage on the y-axis, as position channel works best for showing quantitative trends over time. Each income group, i.e. Low, Lower Middle, Upper Middle and High income, is assigned a distinct colour giving 4 coloured lines, which represent the average enrolment trend of respective groups from 2000 to 2021, making it clear how different economic tiers progressed over time.

Visualisation 1 showed the global wealth-education relationship at country level, and Visualisation 2 determined the cause-effect direction of spending and wealth. None of them determined whether the change in education is consistent across all income groups and regions over time. Visualisation 3 directly answers this by showing whether low income countries are catching up with wealthier nations or falling behind.

The strength of this design is that the convergence or divergence between income groups is immediately visible. If lines move closer, the gap is closing. The weakness is that group averages overlook within-group variation. For example, a fast-improving country like Rwanda is averaged out with other stagnant or declining countries in the same income group (see Fig. 3).

3 PART 3. IMPLEMENTATION

3.1 Data Processing

In the data processing step, the IWI dataset contained both national and subnational rows, and it was filtered to national-level rows only to ensure comparability with the country-level education data. The IWI dataset was then reshaped from wide to long format. The IWI was chosen over GDP as it measures actual household asset ownership, reflecting living standards more directly than national income figures. Secondary enrolment was chosen as the primary education metric because literacy rate had 68% missing data and primary enrolment was almost similar for most countries. Enrolment values above 100% (due to late or re-admissions) were capped as per World Bank data artefact. Additionally, income group classifications were sourced from World Bank population metadata to enable income-based analysis in Q3. Data processing was done in Python and the script is not submitted with this analysis as per coursework instructions. The three datasets were merged on ISO country code and year: IWI Wealth Index [2], World Education Data [6], and World Bank Population Data [5]. The final merged dataset had 1,504 rows covering 126 countries from 2000 to 2021.

3.2 WebApp Live Link

The webapp runs locally via a Python HTTP server. Source code and data files are included in the submission folder.

3.3 Visualisation Implementation

The webApp was implemented as a self-contained HTML file using D3.js v7. The pipeline loads data from the final merged CSV file at runtime and displays it in the form of a bubble chart. Three interaction features were implemented: year slider (from 2000 to 2021), hover tooltips on bubbles, and continent filter. The countries in the dataset were classified into 4 continents, and the chart can be filtered on continents for viewing specific continent data. The year slider at bottom animates bubble positions with 500ms transition, and the tooltip shows the following details: country name, IWI score, enrolment, population, and income group upon hovering. The continent checkboxes filter bubbles with smooth exit transition, and the scaleSqrt function is used for bubble radius so area is proportional to population. This implementation was adapted from the D3 Graph Gallery bubble chart example [3], which was modified to incorporate the wealth-education-population dataset and custom interaction features. Together, the three visualisations explore the wealth-education-population relationship from different angles: Q1 shows the overall correlation, Q2 investigates whether spending drives wealth or vice versa, and Q3 tracks whether poorer countries are closing the education gap over time.

ACKNOWLEDGMENTS

During the research phase of this project, AI tools were used for data exploration, searching relevant population datasets on the internet, brainstorming research questions, and understanding dataset structures. All the written content, visualisation designs, and implementation are the author’s own work [4].

REFERENCES

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[5] World Bank. Population, total (SP.POP.TOTL). World Development Indicators, 2024. Accessed April 2026.

[6] World Bank. World development indicators: Education statistics. Kaggle dataset compiled by Bushra Qurban, 2024. Accessed April 2026.

APPENDIX A: VISUALISATION SKETCHES

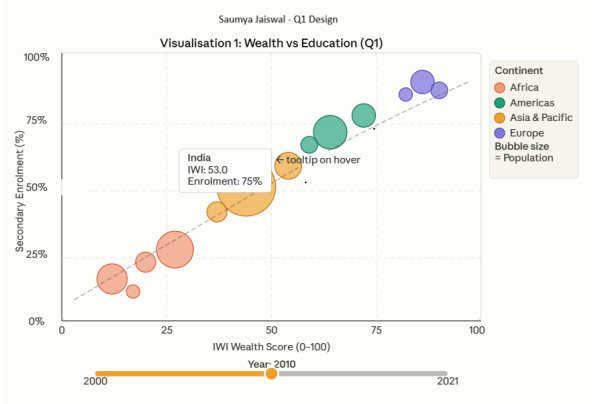


Figure 1: Visualisation 1: Bubble chart, wealth vs. education over time (Q1).

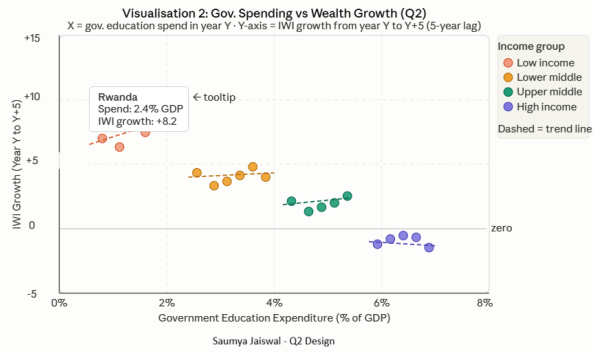


Figure 2: Visualisation 2: Scatter plot, gov. spending vs. wealth growth (Q2).

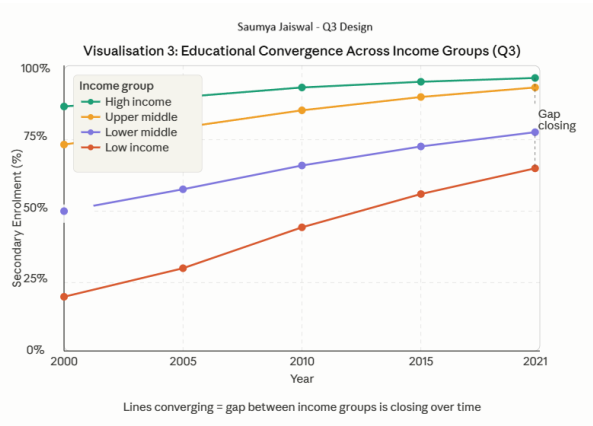


Figure 3: Visualisation 3: Line chart, educational convergence (Q3).